ERA 5 calibration to Elexon power

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2026-03-31

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Overview

Plan

- Literature review
- Calibration methods write-up
  (linear, spatiotemporal & quantile regression)
- Power curve modelling write-up
  (parametric, nonparametric, penalised regression)
- Calibration at aggregate vs. site level
- Cut-off estimation
- Wind drought extremes

Calibration of generic power curve estimate

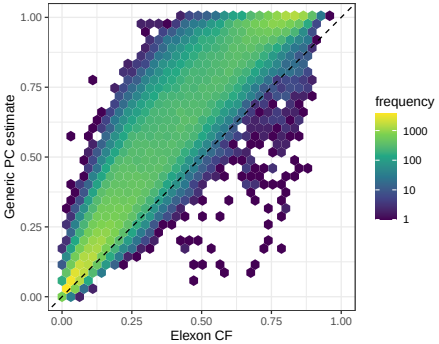
GB level aggregation

- Nearest ERA5 grid for wind speed
- Vertical interpolation using a power law
- Generic Power Curve (onshore, offshore)
- Normalised ower aggregated at GB level

$$p_{\text{tot},t} = \sum_i p_{it} / \sum_i C_i$$

- Hourly resolution
- Potential generation (ex curtailment)

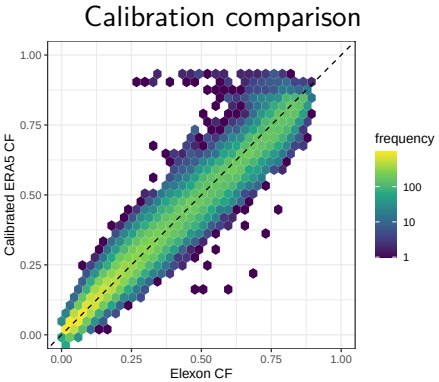
Scatter aggregated power



Linear model

- Using step AIC criteria selection:
  - k: Type (offshore / onshore)
  - m: month, h: hour
  - w_t: wind speed

$$p_t^{calib} = \alpha_{k,m,h} + \beta_{k,m}\hat{p}_t + \sum_{j=1}^{3} \phi_j w_t^3$$

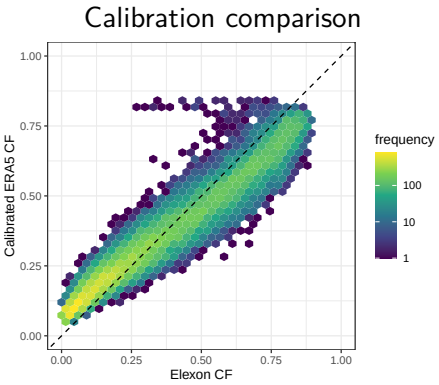


Autoregressive error

- Using step AIC criteria selection:
  - k: Type (offshore / onshore)
  - m: month, h: hour
  - w_t: wind speed
  - u_t: AR1

$$p_t^{calib} = \alpha_0 + \sum \alpha_e + \beta_k \hat{p}_t + f_w(w_t) + u_t$$

where α_0 is the intercept, the α_e $e \in k, m, h$ represent random effects (cyclic for month and hour), β_k is a random slope, f_w is smooth effect (RW2).

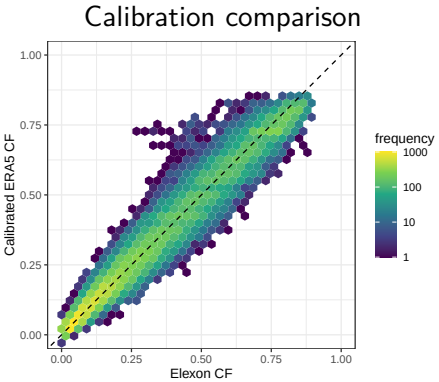


Hierarchical Bayesian model

- Turning muliplicative corrections into smooth functions

$$p_t^{calib} = \alpha + \beta_k + \beta_m + \beta_h + s_p(p_t) + s_w(w_t)$$

where β_k, β_m, β_h are random effects and s_p and s_w are smooth functions

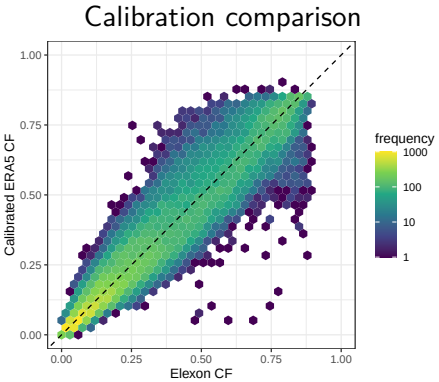


Quantile Mapping

- Quantile mapping aims to replicate observed distribution

$$p_t^{calib} = F_{\text{obs}}(F_{\text{mod}}^{-1}(p_t))$$

where F_{mod} and F_{obs} are the CDF of the power estimate and observed values



Comparison scores

model	RMSE	MAE	Bias
Generic PC	0.200	0.164	1.63e-01
AR inlabru	0.092	0.077	3.16e-04
AR1	0.075	0.062	8.57e-05
QM	0.075	0.054	-2.60e-05
Linear	0.052	0.038	1.39e-19
Linear inlabru	0.049	0.036	-2.61e-10

Power curve modelling

Power curve modelling

Components

- Power curve estimate from data
- Probability of zero generation
- Penalisation to make power curve resemble manufaturer's PC

Using a three-likelihood approach for power curve estimation:

- **1st likelihood:** Observed power is modelled as a smooth function of wind speed
- **2nd likelihood:** Zero inflated component
- **3rd likelihood** Penalisation towards a generic PC

Power model overview

1. $\pi_{it} = \pi(w, i) = \Pr(P_{it} = 0)$ depends of wind speed and location

$$L^{(0)}(\theta^0|\mathbf{p}, \mathbf{w}, \mathbf{i}) = \prod_t \left[\pi_i(w_{it}, i)^{\mathbb{1}(P_{it}=0)} \times (1 - \pi_i(w_{it}, i))^{\mathbb{1}(P_{it}>0)}\right]$$

2. Generation when $P_{it} > 0$ also depends on wind speed / location. Learned as a smooth function from data: $s^{\mathrm{pc}}(w, i) \in (0, 1)$

$$s^{\mathrm{pc}}(w_{it}, i) = \mathsf{E}(P_{it}|P_{it} > 0, w_{it}, i) = \mu^+_{it}$$
$$L^{(\beta)}(\theta^{\mathrm{pc}}|\mathbf{p}, \mathbf{w}, \mathbf{i}) = \prod_{t:P_{it}>0} \mathsf{Beta}\left(P_{it}|\mu^+_{it}, \phi\right),$$

where θ^0 and θ^{pc} represents the parameters of the smooth functions, $\pi(w, i)$ and s^{pc}, and ϕ the precision of the Beta distribution.

Penalisation component

3. Finally, we penalise s^{pc} to be close to our generic power curve estimate, $\hat{PC}_i(w)$: $d_{it} = |\hat{PC}_i(w_{it}) - s^{\mathrm{pc}}(w_{it}, i)|$ should be small.

- Or maximise the negative of $\tau_{\mathrm{ps}} \sum d_{it}^2$ for a certain penalisation parameter τ_{ps}.
- Adding that penalty term is similar to assume $d_{it} \sim N(0, 1/\tau_{\mathrm{ps}}^{-1})$

$$\max_{\theta} \quad l^{(0)}(\theta) + l^{(\beta)}(\theta) - \tau_{\mathrm{ps}} \sum_{i,t} d_{it}^2$$

where $\theta = (\theta^0, \theta^{\mathrm{pc}}, \tau_{\mathrm{ps}})$ represents the parameters of both likelihoods, plus the penalisation parameter.

Power curve model equations

Model is a Zero Inflated Beta (ZIB) with penalisation component:

Zero–probability model (Bernoulli)

$$\eta_{it}^{(0)} = \alpha_{\text{zero}} + f_{\text{zero}}(w_{it},\, i)\,,$$

$$Z_{it} \mid \eta_{it}^{(0)} \sim \text{Bernoulli}(\pi_{it}), \qquad \pi_{it} = \frac{1}{1+\exp\left\{-\eta_{it}^{(0)}\right\}}\,,$$

Positive–output model (Beta)

$$\eta_{it}^{(\beta)} = \alpha_{\text{pc}} + f_{\text{pc}}(w_{it},\, i)\,,$$

$$P_{it}^{\text{obs}} \mid \eta_{it}^{(\beta)} \sim \text{Beta}(\mu_{it}^{+}, \phi), \qquad \mu_{it}^{+} = \frac{1}{1+\exp\left\{-\eta_{it}^{(\beta)}\right\}}\,,$$

Pseudo–likelihood model (Gaussian)

$$\hat{PC}_i(w_{it}) \mid \eta_{it}^{(\beta)} \sim \mathcal{N}\left(\eta_{it}^{(\beta)},\, \tau_{\text{ps}}^{-1}\right), \qquad \tau_{\text{ps}} = \text{precision or penalisation}\,.$$

Details on smooth components

- The smooth components $f_{\text{zero}}(w, i)$ $f_{\text{pc}}(w, i)$ is modelled as a 1D SPDE over wind speed with a uniform correlation matrix across locations $i = 1, \dots, I$.

$$f(w, i) = \sum_{i=1}^{I} \beta_i \psi_k(w),$$

where $\psi_k(w)$ are basis functions defined over wind speed and β_i are coefficients The correlation structure of $f(w, i)$ assumes a Matern covariance function: Let w_s, w_h represent two wind speed values in location i, and $d = |w_s - w_t|$ their distance, then:

$$\begin{aligned}\text{Cor}(d) &= \text{Cor}(f(w_s, i), f(w_t, i)) \\ &= \frac{1}{2^{\nu-1}\Gamma(\nu)}(\kappa d)^{\nu} K_{\nu}(\kappa d), \quad \alpha = \nu + d/2,\end{aligned}$$

Details on smooth components

- Additionally a uniform correlation matrix is assumed across locations $j, k \in \{1, \dots, I\}$. That is:

$$\mathrm{Cor}(f(w, j), f(w, k)) = \rho, \quad j \neq k$$

which helps to pool information across locations, benefiting locations with less data.

Power estimate

This leads to the following power estimate at time t and location i:

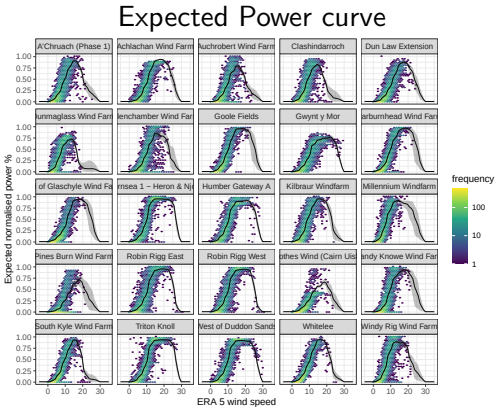
$$\hat{P}_{it} = \begin{cases} 0, & \text{with probability } \pi_{it}, \\ \mu_{it}^{+}, & \text{with probability } 1 - \pi_{it}. \end{cases}$$

Which leads to the expected power estimate:

$$E[P_{it}|w_{it}, i] = (1 - \pi_{it}) \times \mu_{it}^{+}$$

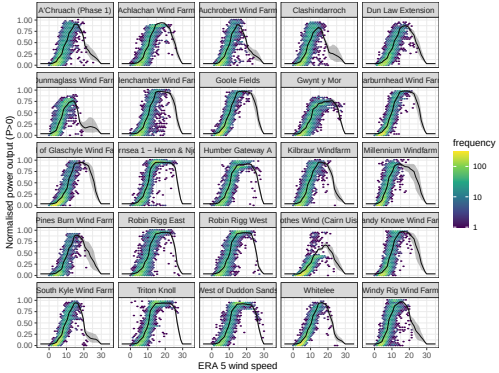
Power curve model estimates

- Starting models with
  - 1Y of data (2024)
  - 25 wind farms
  - Hourly data
  - Excluding outages listed in REMIT

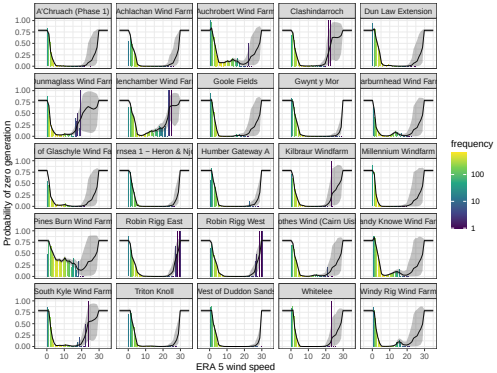


Power curve model estimates

Power curve estimate



Probability of zero generation

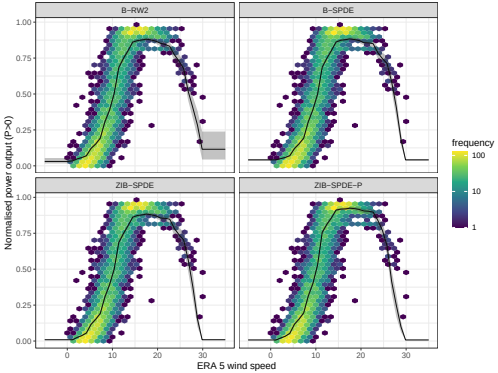


Alternative versions

- Beta model with RW2 power curve (B-RW2)
- Beta model with 1D SPDE power curve (B-SPDE)
- ZIB model with 1D SPDE power curve (ZIB-SPDE)
- ZIB model with 1D SPDE power curve penalised (ZIB-SPDE-P)

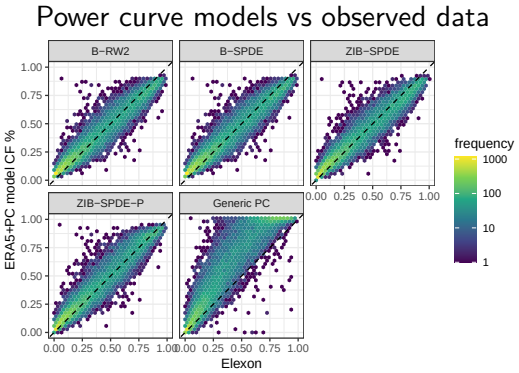
Zero Inflated models better capture low wind power regions.

Power curve estimates under different models



ZIB model vs observed data

- Power curve modelling highly improves the calibration
- ZIB models deal better with the tail regions
- Penalised version actually pulls estimate toward Generic PC
- We could pull estimate instead to a 5 parameter curve estimate



Error metrics

Error metrics calculated on 30% of the sampled wind farms data.

model	RMSE	MAE	Bias
Generic PC	0.2188	0.1583	0.1469
B-RW2	0.0995	0.0723	0.0191
B-SPDE	0.0994	0.0722	0.0190
ZIB-SPDE	0.0976	0.0691	0.0103
ZIB-SPDE-P	0.1016	0.0694	0.0269

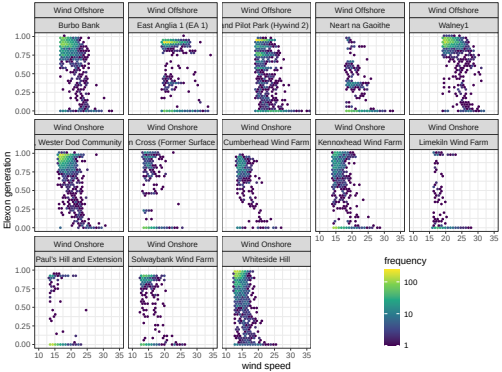
Pooling data for cut-off estimation

- Not enough data to estimate at wind farm level
- Even using all 5 years of data (2019-2024), only a few farms show cut-off behaviour

Pooling data at high wind speeds

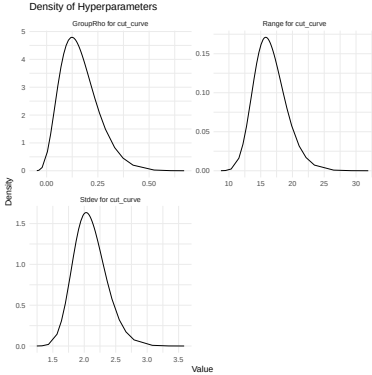
wind speed	prop. P=0	prop. P<=0.02	n obs.
[0,15]	0.22	0.23	3292
(15,20]	0.13	0.14	7062
(20,25]	0.11	0.13	5266
(25,30]	0.43	0.48	670
(30,Inf]	0.76	0.79	63

Tail behaviour of power curve

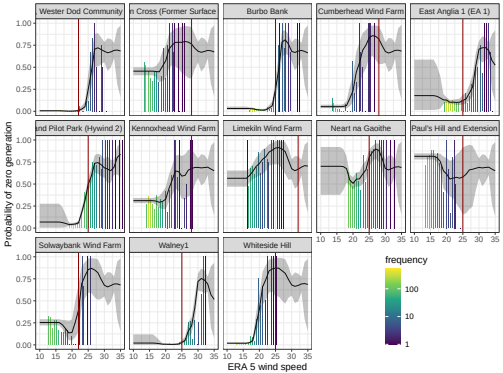


Model to estimate cut-off by wind speed

Hyperparameters of the cut-off model



Probability of zero generation at high wind speeds



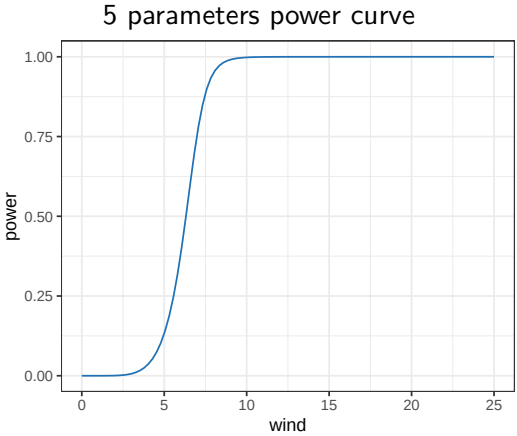
Power Curve fitting

Classic parametric shape

The five-parameter formula (Lydia et.al, 2013):

$$P(u) = D + \frac{A - D}{(1 + (u/C)^B)^G}$$

where: A and D are the upper and lower asymptotes, C controls the inflection point, B is related to the slope at inflection point, and G controls the asymmetry.



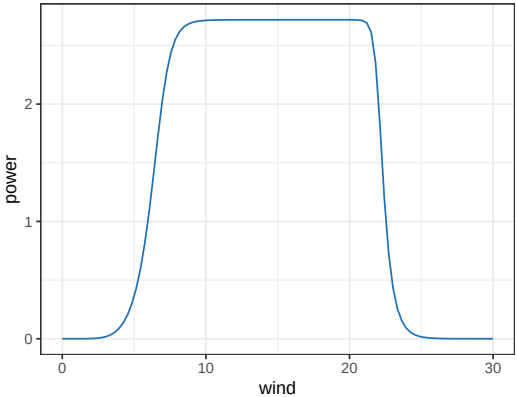
Parametric shape for cut-off

Adding extra parameters we can model de decay after cut-off:

$$P(u) = D + \frac{A - D}{(1 + (u/C_1)^{B_1})^G (1 + (u/C_2)^{B_2})^G}$$

where: A and D are the upper and lower asymptotes, C_1 is the first inflection point, C_2 is related to the cut-off, B_1 is related to the slope at inflection point, B_2 to the slope during cut-off, and G controls the asymmetry.

7 parameters power curve

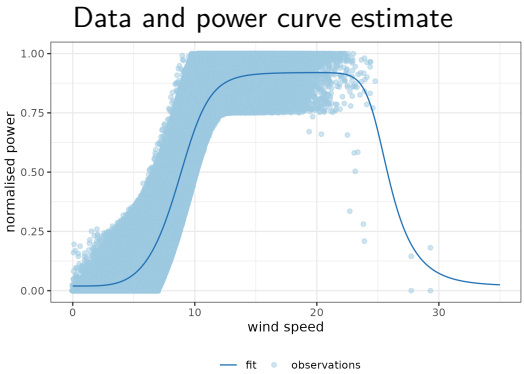


Parametric fit for Scottish wind farms

- We can fit the parametric shape using optim
- As long as there is a reasonable amount of points showing the cut-off optim can converge

Estimates from data

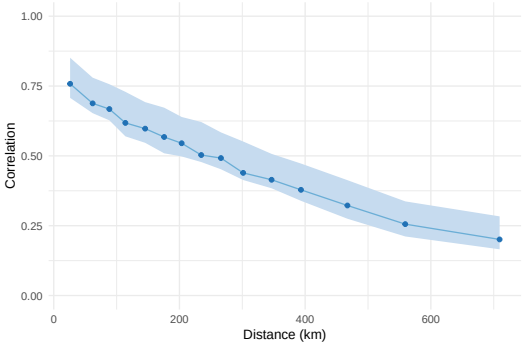
par	estimate
A	0.92
B1	-9.74
C1	10.06
D	0.02
G	0.43
B2	35.64
C2	24.92



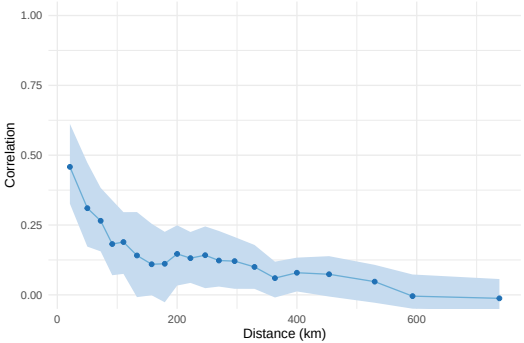
Spatial correlation

Spatial correlation

Correlation for Elexon generation



Correlation for error after calibration



Next Steps

- Add CRPS and coverage metrics
- Spatiotemporal calibration model
- Write introduction and methods

Appendix

Data sources

Wind speed

- ERA5 at wind farms
  - Hourly data
  - Spatial resolution 0.25° x 0.25°
  - 10m and 100m heights

Wind power

- Elexon BMU data (since 2019)
  - Half hourly data
  - Generation, curtailment, potential, capacity
  - Outage data (REMIT)
- REPD database
  - Location, turbine height, capacity

Previous research

Power curve modelling

Binning method

$$P_i = \frac{1}{n_i} \sum_{j=1}^{n_i} P_{ij}$$

where: P_{ij} is the j th power observation in bin i and n_i no. of observations in bin i

Logistic

$$P(u) = a \frac{1 + m \exp(-u/\tau)}{1 + n \exp(-u/\tau)}$$

where a represents the upper asymptote, n, m shape the lower asymptote, and τ controls the transition.

Power curve modelling

5 parameter curve

$$P(u) = D + \frac{A - D}{(1 + (u/C)^B)^G}$$

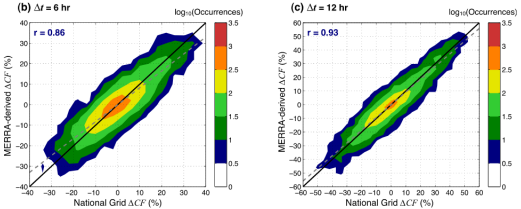
where: A and D are the upper and lower asymptotes, C is the inflection point, B the slope at inflection point, adn G controls the asymmetry.

Reanalysis data to quantify extreme wind power statistics

D. Cannon, D. Brayshaw, et. al (2015)

- MERRA wind speed validated with MIDAS
- Vertical interpolation with a logarithmic change
- Calibrated power curves based on manufacturers PC
- Use that to analys extreme low and high levels, and ramps

ΔCF comparison

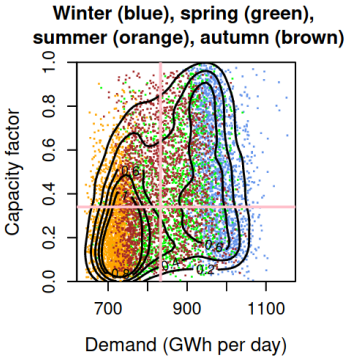


Balancing energy

H. Thornton, D. Brayshaw (2020) analyse the relationship between weather, energy demand, and wind power.

- ERA Interim wind speed
- Cubic power curve with air density correction
- CF compared with GB average from other studies
- Seasonal effects on Demand and

ΔCF comparison

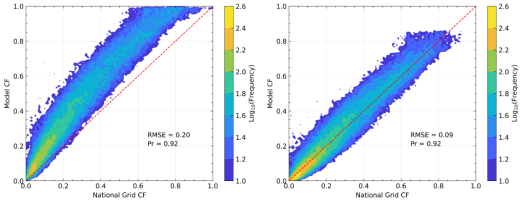


Analysis of extreme wind droughts

Panit Potisomporn, C. Vogel (2024) perform a extreme value analysis of wind droughts in GB.

- Use ERA5 wind speeds calibrated to MIDAS with QM
- Build a ML algorithm that learns how to extrapolate wind speed from 10m to hub height
- Use a 5 parameter logistic function to model power curve
- Model energy losses with factors by type.

CF calibration



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